



GIRRAWEE HIGH SCHOOL

YEAR 11 – Task 2

2009

MATHEMATICS

Time allowed – 90 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a *new* sheet of paper.

QUESTION 1 (18 marks)

Marks

- (a) Evaluate $\frac{7.63 - 4.82}{\sqrt{(8.2)^2 - 58.7}}$ correct to 3 decimal places. **2**
- (b) Write 0.00471825 in scientific notation correct to 3 significant figures. **2**
- (c) Evaluate $|-3| + |-2|^2 - |4 \times -5|$. **2**
- (d) Express $0.\dot{3}2\dot{4}$ as a fraction in lowest terms. **3**
- (e) Given that $L = \frac{g}{4} \left(\frac{T}{\pi} \right)^2$, find the value of T when $L = 1.5$, $g = 9.8$ and $T > 0$. Give answer correct to 2 decimal places. **3**
- (f) Find the value of x if $\sqrt{52} + \sqrt{13} = \sqrt{x}$ **3**
- (g) After GST of 10% the new price is \$12.50.
What was the price before GST?
(Round the answer to the nearest 10 cents). **3**

QUESTION 2 (23 marks)

Marks

(a) Simplify:

(i) $\sqrt{700}$

2

(ii) $\sqrt{8} + \sqrt{32} - \sqrt{50}$

3

(b) Expand and simplify:

(i) $(2\sqrt{5} - 1)(3\sqrt{5} + 4)$

3

(ii) $(2\sqrt{5} - 3)^2$

3

(c) Express with a rational denominator :

(i) $\frac{2}{3\sqrt{5}}$

2

(ii) $\frac{6}{\sqrt{2} + \sqrt{5}}$

3

(d) Solve for x :

(i) $|x - 3| = 7 - 2x$

4

(ii) $|3x - 4| \leq 2$

3

QUESTION 3 (21 marks)**Marks**

(a) Expand and simplify $(4x + 3y)(4x - 3y) - (5x - 3y)^2$.

3

(b) Factorise :

(i) $16m^2 - 9$

2

(ii) $x^2 - 2x - 63$

2

(iii) $3a^2 - 10a + 8$

2

(iv) $27 - y^3$

3

(v) $2ax + 6a - 8x - 24$.

3

(c) Express with positive indices $a^{-3} \div 4a^{-4} \times 3a^5$.

3

(d) Find the values of x and y if $\frac{6}{\sqrt{3}+2} = x + y\sqrt{3}$.

3**QUESTION 4 (17 marks)**

(a) Simplify:

(i) $m + \frac{3}{m-1}$

2

(ii) $\frac{7x-1}{4} - \frac{x+2}{3}$

3

(b) Simplify:

(i) $\frac{3m^2 + 24}{12 - 3m^2}$

2

(ii) $\frac{x^2 + 3x}{x-2} \times \frac{x^2 - 6x + 8}{x^2 - 4x}$

4

(c) Solve:

(i) $\frac{4x}{x+3} = \frac{2}{3}$

3

(ii) $\frac{1}{3}(m+1) - \frac{1}{4}(m-2) = 2$

3

QUESTION 5 (26 marks)

Marks

(a) Sketch the following graphs:

(i) $y = |x| + 1$ 2

(ii) $y = \frac{1}{x-2}$ 2

(iii) $x^2 + y^2 = 4$ 2

(iv) $y = 2^x$ 2

(b) Which of the above graphs does not represent a function? 1

(c) State the domain and range of the following functions:

(i) $y = x^2$ 2

(ii) $y = \sqrt{4 - x^2}$ 2

(iii) $y = \frac{1}{x-2}$ 2

(d) A function is defined by the rule 2

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ -1 & \text{if } 0 < x < 2 \\ x & \text{if } x \geq 2. \end{cases}$$

Evaluate $f(-1) + f(1) + f(5)$.

(e) Determine whether the function $y = 8x^3 - 7x - 7$ is even, odd, or neither. Justify your answer. 3

(f) Shade in the region on the number plane for which $x - 2y > 0$ and $2x + y > 0$ hold simultaneously. 3

(g) Find the centre and radius for the circle whose equation is $x^2 + y^2 - 4x - 14y + 17 = 0$. 3

QUESTION 6 (18 marks)

Marks

(a) Evaluate $\cos 58^\circ 19'$ correct to 3 decimal places.

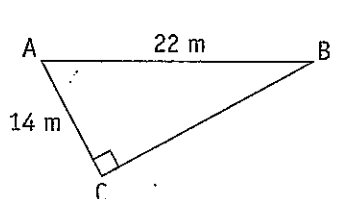
2

(b) Find the value of θ in degrees and minutes if $\tan \theta = 0.348$

2

(c) Find the value of $\angle BAC$ in degrees and minutes to the nearest minute.

3



(d) Find the exact value of $\tan 300^\circ$.

2

(e) Simplify $\sin B \cot B$.

2

(f) Prove that $\frac{1}{1 - \sin \theta} - \frac{1}{1 + \sin \theta} = \frac{2 \sin \theta}{\cos^2 \theta}$.

3

(g) Solve for $0^\circ \leq \theta \leq 360^\circ$

(i) $2 \cos \theta = \sqrt{3}$

2

(ii) $\tan \theta + 1 = 0$

2

QUESTION 7 (13 marks)

Marks

- (a) Amberville (A) is 35km from Bordernook (B) and on a bearing of 245° .
Coopertown (C) is due south of Bordernook and on a bearing of 155° from Amberville.

(i) Show this information on a diagram.

3

(ii) Find the size of $\angle BAC$.

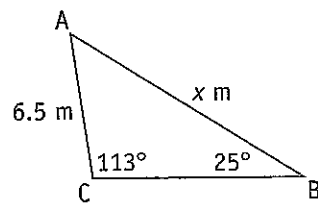
2

(iii) How far is A from C? (Give answer to 1 decimal place).

2

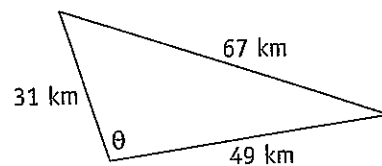
- (b) Use the sine rule to find the value of x .
Give answer to 1 decimal place.

3



- (c) Use the cosine rule to find the value of θ to the nearest whole degree.

3



END OF TASK

Year 11, Mathematics, Task 2, 2009 Solutions

Question 1 (18 marks)

$$(a) \frac{7.63 - 4.82}{\sqrt{8.2^2 - 58.7}} = 0.962 \quad (2)$$

$$(b) 0.00471825 = 4.72 \times 10^{-3} \quad (2)$$

$$(c) |-3| + |-2|^2 - |4 \times -5| = 3 + 4 - 20 = -13. \quad (2)$$

$$(d) \text{Let } x = 0.324324... \quad (1)$$

$$\therefore 1000x = 324.324324... \quad (2)$$

$$(2) - (1) \text{ gives } 999x = 324$$

$$\therefore x = \frac{324}{999}$$

$$x = \frac{12}{37} \quad (3)$$

$$(e) L = \frac{9}{4} \left(\frac{T}{\pi} \right)^2$$

$$\therefore 1.5 = \frac{9.8}{4} \left(\frac{T}{\pi} \right)^2$$

$$\frac{6}{9.8} = \frac{T^2}{\pi^2}$$

$$\therefore T^2 = \frac{6\pi^2}{9.8} \quad (3)$$

$$\therefore T = \pm 2.46$$

$$\therefore T = 2.46 \text{ since } T > 0$$

$$(f) \sqrt{52} + \sqrt{13} = \sqrt{x}$$

$$2\sqrt{13} + \sqrt{13} = \sqrt{x}$$

$$3\sqrt{13} = \sqrt{x}$$

$$\therefore \sqrt{117} = \sqrt{x}$$

$$\therefore x = 117. \quad (3)$$

$$(g) 110\% = \$12.50$$

$$\therefore 100\% = \frac{\$12.50}{110} \times 100$$

$$= \$11.36$$

$$\therefore \$11.40. \quad (3)$$

Question 2 (23 marks)

$$(a) (i) \sqrt{700} = \sqrt{100 \times 7} = 10\sqrt{7}. \quad (2)$$

$$(ii) \sqrt{8} + \sqrt{32} - \sqrt{50} = 2\sqrt{2} + 4\sqrt{2} - 5\sqrt{2} = \sqrt{2} \quad (3)$$

$$(b) (i) (2\sqrt{5} - 1)(3\sqrt{5} + 4) = 6 \times 5 + 8\sqrt{5} - 3\sqrt{5} - 4 = 26 + 5\sqrt{5}. \quad (3)$$

$$(ii) (2\sqrt{5} - 3)^2 = 4 \times 5 - 12\sqrt{5} + 9 = 29 - 12\sqrt{5}. \quad (3)$$

$$(c) (i) \frac{2}{3\sqrt{5}} = \frac{2}{3\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{2\sqrt{5}}{15}. \quad (2)$$

$$(ii) \frac{6}{\sqrt{2} + \sqrt{5}} = \frac{6}{\sqrt{2} + \sqrt{5}} \times \frac{\sqrt{2} - \sqrt{5}}{\sqrt{2} - \sqrt{5}} = \frac{6(\sqrt{2} - \sqrt{5})}{2 - 5} = -2(\sqrt{2} - \sqrt{5}) \quad (3)$$

(d)(i) $|x-3| = 7-2x$

$x-3 = 7-2x$ or $-(x-3) = 7-2x$

$3x = 10$

$-x+3 = 7-2x$

$\therefore x = \frac{10}{3}$

$x = 4$

Check: For $x = \frac{10}{3}$,

LHS = $|\frac{10}{3}-3| = \frac{1}{3}$

RHS = $7-2 \times \frac{10}{3} = \frac{1}{3}$

For $x=4$,

LHS = $|4-3| = 1$.

RHS = $7-2 \times 4 = -1$.

$\therefore x = \frac{10}{3}$ or $3\frac{1}{3}$ is the only solution.

(ii) $|3x-4| \leq 2$

$\therefore -2 \leq 3x-4 \leq 2$

$-2+4 \leq 3x \leq 2+4$

$2 \leq 3x \leq 6$

$\therefore \frac{2}{3} \leq x \leq 2$

Question 3 (21 marks)

(a) $(4x+3y)(4x-3y) - (5x-3y)^2$

$= 16x^2 - 9y^2 - (25x^2 - 30xy + 9y^2)$

$= 16x^2 - 9y^2 - 25x^2 + 30xy - 9y^2$

$= -9x^2 - 18y^2 + 30xy$

(b) (i) $16m^2 - 9$

$= (4m-3)(4m+3)$

(ii) $x^2 - 2x - 63$

$= (x+7)(x-9)$

(iii) $3a^2 - 10a + 8$

$= (3a-4)(a-2)$

(iv) $27 - y^3$

$= 3^3 - y^3$

$= (3-y)(9+3y+y^2)$

(v) $2ax + 6a - 8x - 24$

$= 2(ax + 3a - 4x - 12)$

$= 2[a(x+3) - 4(x+3)]$

$= 2(a-4)(x+3)$

(c) $a^{-3} \div 4a^{-4} \times 3a^5$

$= \frac{a^{-3}}{4a^{-4}} \times 3a^5$

$= \frac{1}{4a^{-1}} \times 3a^5 = \frac{3}{4}a^6$

(d) $\frac{6}{\sqrt{3}+2} \times \frac{\sqrt{3}-2}{\sqrt{3}-2}$

$= \frac{6(\sqrt{3}-2)}{3-4} = -6(\sqrt{3}-2)$

$= -6\sqrt{3} + 12$

$= y\sqrt{3} + x$

$\therefore x = 12, y = -6$

Question 4 (17 marks)

(a) (i) $m + \frac{3}{m-1}$

$= \frac{m(m-1)+3}{m-1}$

$= \frac{m^2 - m + 3}{m-1}$

(ii) $\frac{7x-1}{4} - \frac{x+2}{3}$

$= \frac{3(7x-1) - 4(x+2)}{12}$

$= \frac{21x - 3 - 4x - 8}{12}$

$= \frac{17x - 11}{12}$

(b) (i) $\frac{3m^2+24}{12-3m^2}$

$= \frac{3(m^2+8)}{3(4-m^2)}$

$= \frac{m^2+8}{4-m^2}$

(2)

(ii) $\frac{x^2+3x}{x-2} \times \frac{x^2-6x+8}{x^2-4x}$

$= \frac{x(x+3)}{x-2} \times \frac{(x-2)(x-4)}{x(x-4)}$

$= x+3$

(4)

(c) (i) $\frac{4x}{x+3} = \frac{2}{3}$

$\therefore 12x = 2(x+3)$

$12x = 2x + 6$

$\therefore 10x = 6$

$\therefore x = \frac{3}{5}$

(3)

(ii) $\frac{1}{3}(m+1) - \frac{1}{4}(m-2) = 2$

Multiply by 12 throughout,

$\therefore 4(m+1) - 3(m-2) = 24$

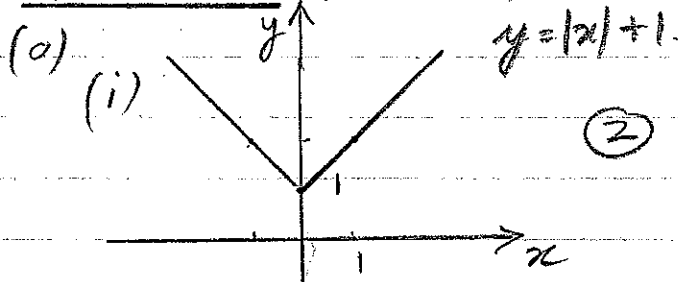
$4m+4-3m+6 = 24$

$m+10 = 24$

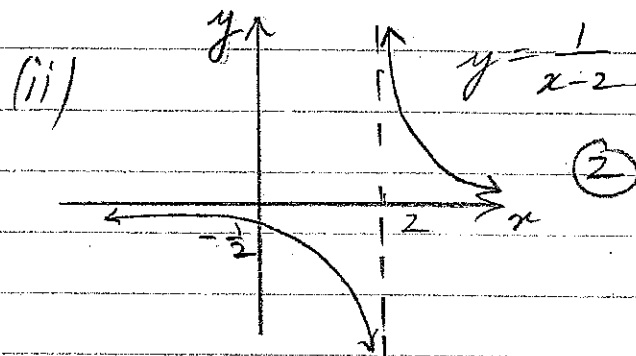
$\therefore m = 14$

(3)

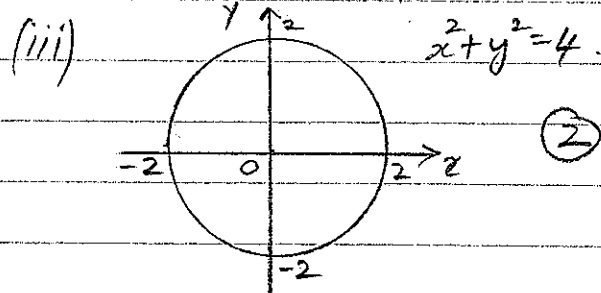
Question 5 (26 marks)



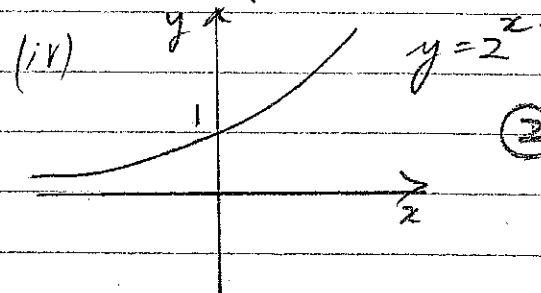
(2)



(2)



(2)



(2)

(b) (iii) above

(1)

(c) (i) $y = x^2$

D: all real x

R: $y \geq 0$

(2)

(ii) $y = \sqrt{4-x^2}$

D: $-2 \leq x \leq 2$

R: $0 \leq y \leq 2$

(2)

(iii) $y = \frac{1}{x-2}$

D: all real x , $x \neq 2$

R: all real y , $y \neq 0$

(2)

(d) $f(-1) + f(1) + f(5)$

$= 0 + -1 + 5$

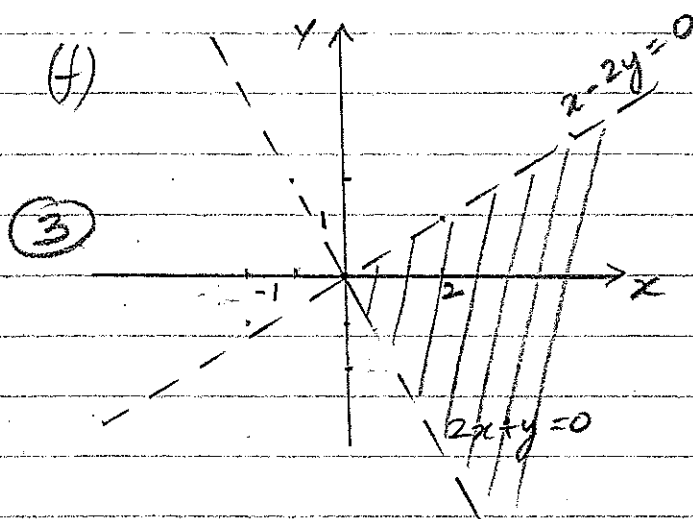
$= 4$

(2)

(e) $f(x) = 8x^3 - 7x - 7$
 $f(-x) = 8(-x)^3 - 7(-x) - 7$
 $= -8x^3 + 7x - 7$

$-f(x) = -8x^3 + 7x + 7$ (3)

Since $f(x) \neq -f(x)$ not even, and $-f(x) \neq f(-x)$ not odd, $\therefore$ neither.



(g) $x^2 - 4x + y^2 - 14y = -17$
 $(x^2 - 4x + 4) + (y^2 - 14y + 49) = -17 + 53$
 $(x-2)^2 + (y-7)^2 = 36$
 $\therefore$ centre (2, 7)
 radius = 6. (3)

Question 6 (18 marks)

(a) $\cos 58^\circ 19' = 0.525$ (2)

(b) $\tan \theta = 0.348$
 $\therefore \theta = 19^\circ 11'$ (2)

(c) $\cos \theta = \frac{14}{22}$
 $\therefore \theta = 50^\circ 29'$ (3)

(d) $\tan 300^\circ$
 $= \tan(360 - 60)$
 $= -\tan 60^\circ = -\sqrt{3}$ (2)

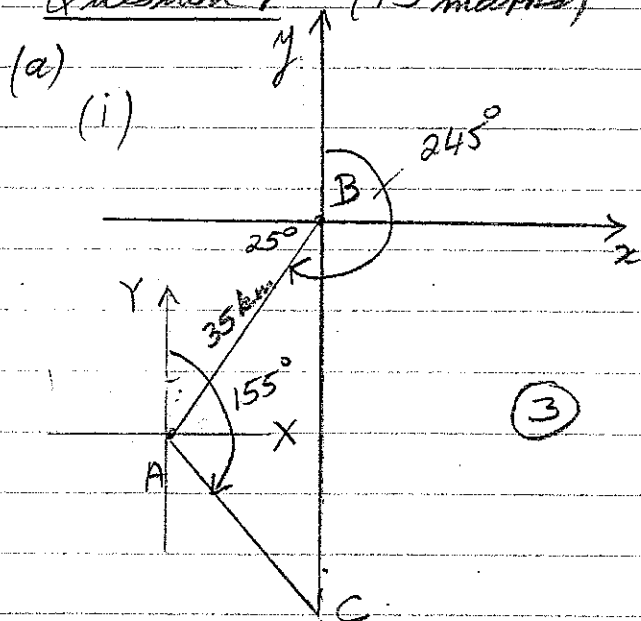
(e) $\sin B \cot B$
 $= \sin B \times \frac{\cos B}{\sin B}$ (2)
 $= \cos B$

(f) LHS = $\frac{1}{1 - \sin \theta} - \frac{1}{1 + \sin \theta}$
 $= \frac{1 + \sin \theta - 1 + \sin \theta}{(1 - \sin \theta)(1 + \sin \theta)}$
 $= \frac{2 \sin \theta}{1 - \sin^2 \theta}$
 $= \frac{2 \sin \theta}{\cos^2 \theta}$ (3)
 $= \text{RHS}$

(g) (i) $2 \cos \theta = \sqrt{3}$ S/A
 $\therefore \cos \theta = \frac{\sqrt{3}}{2}$ T/C
 $\therefore \theta = 30^\circ, 330^\circ$ (2)

(ii) $\tan \theta + 1 = 0$
 $\therefore \tan \theta = -1$ (2)
 $\therefore \theta = 135^\circ, 315^\circ$

Question 7 (13 marks)



$$(ii) \angle ABC = 245^\circ - 180^\circ \\ = 65^\circ$$

$$\angle BAX = 25^\circ, \therefore \angle BAY = 65^\circ$$

$$\therefore \angle XAC = 155^\circ - 90^\circ = 65^\circ$$

$$\therefore \angle BAC = \angle BAX + \angle XAC \\ = 25^\circ + 65^\circ \\ = 90^\circ \quad (2)$$

(iii)

$$\tan 65^\circ = \frac{AC}{35}$$

$$\therefore AC = 35 \tan 65^\circ \\ = 75.1 \text{ km} \quad (2)$$

$$(b) \frac{x}{\sin 113^\circ} = \frac{6.5}{\sin 25^\circ}$$

$$\therefore x = \frac{6.5}{\sin 25^\circ} \times \sin 113^\circ$$

$$x = 14.2 \quad (3)$$

$$(c) 67^2 = 31^2 + 49^2 - 2 \times 31 \times 49 \cos \theta$$

$$\therefore \cos \theta = \frac{31^2 + 49^2 - 67^2}{2 \times 31 \times 49}$$

$$= \frac{-1127}{3038}$$

$$\therefore \theta = 111.775^\circ \quad (3) \\ = 112^\circ$$